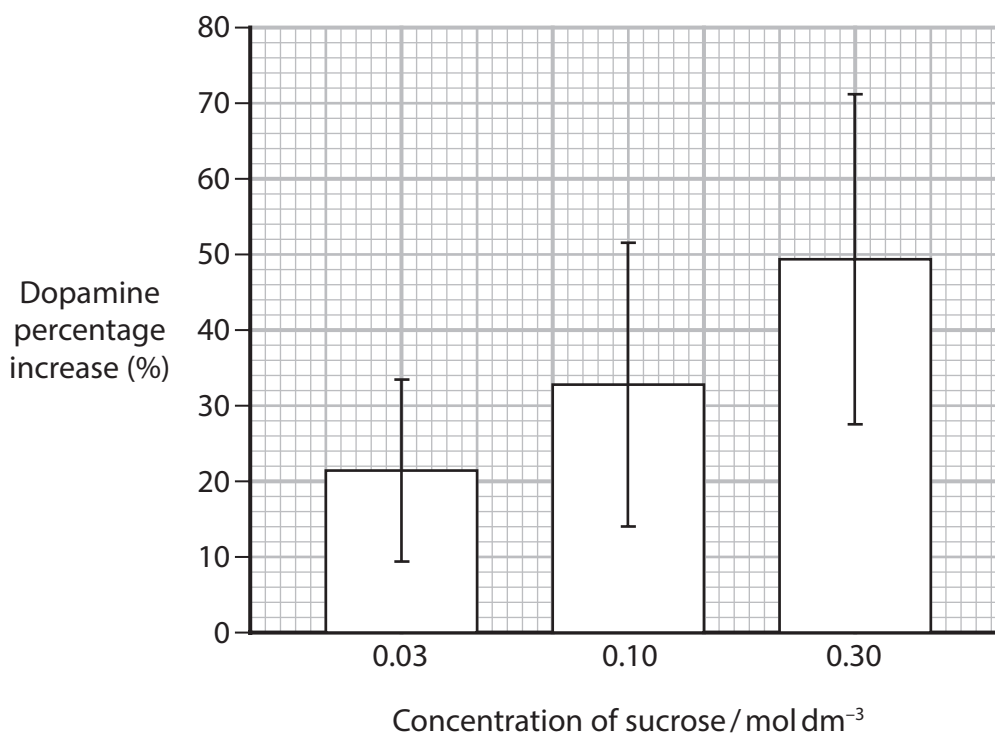


**7** (a) Scientists investigated the effect of sucrose solutions on the desire of rats to eat.

The release of dopamine in some areas of the brain increases the desire of the rat to eat.

Scientists measured increases in the release of dopamine in the brain of rats given different concentrations of sucrose solutions to drink.



The scientists concluded that drinking a sucrose solution had an effect on the desire of the rats to eat.

- (i) Comment on the results shown in the bar chart and how they support this conclusion.

(4)

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(ii) The table gives the results for a sucrose concentration of  $0.4 \text{ mol dm}^{-3}$ .

Complete the table to calculate the standard deviation ( $S$ ) for these results.

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Give your answer to two significant figures.

(2)

| result ( $x$ ) | $x - \bar{x}$            | $(x - \bar{x})^2$ |
|----------------|--------------------------|-------------------|
| 45             | -15                      | 225               |
| 63             | 3                        | 9                 |
| 74             | 14                       | 196               |
| 58             |                          |                   |
|                | $\sum (x - \bar{x})^2 =$ |                   |
|                | $n - 1 =$                |                   |
|                | $S =$                    |                   |

(b) When humans are in danger or under stress they respond by activating the 'fight or flight' response.

Describe the role of adrenaline in the fight or flight response.

(3)

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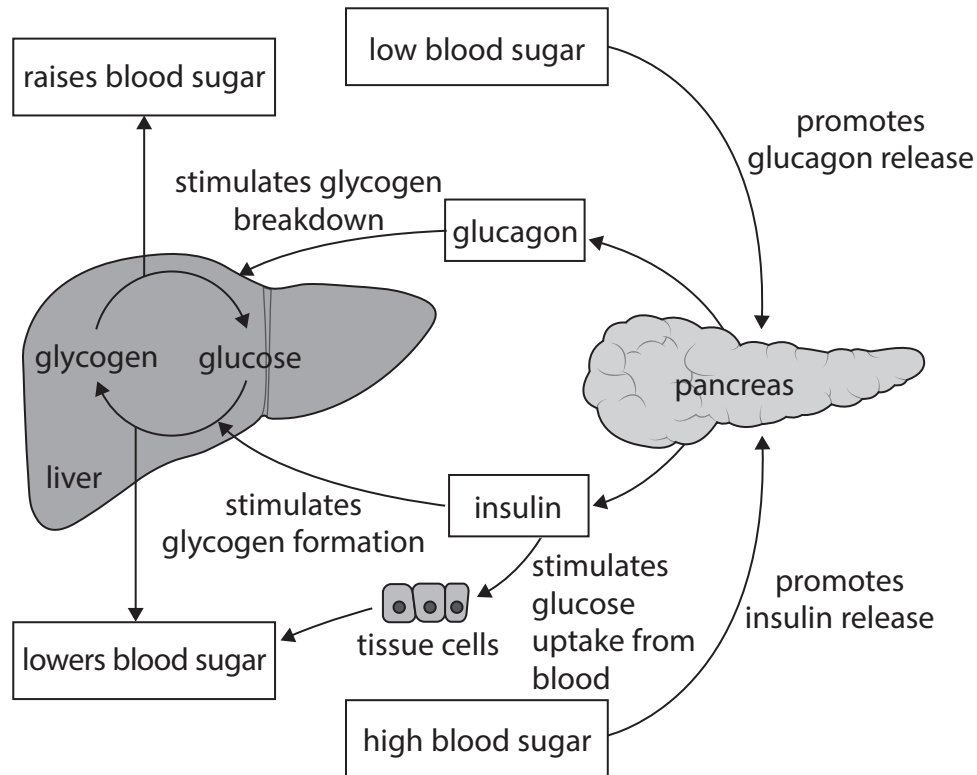
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\*(c) Negative feedback is involved in maintaining systems within narrow limits.

The blood glucose concentration has to be maintained within narrow limits.

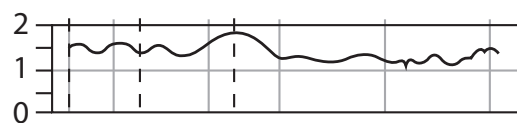
Insulin and glucagon are hormones involved in the control of blood glucose levels.

The diagram shows how blood glucose concentrations are controlled in the body.

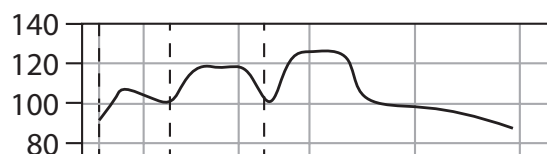


The graphs show the daily changes in the concentrations of insulin, glucagon and glucose.

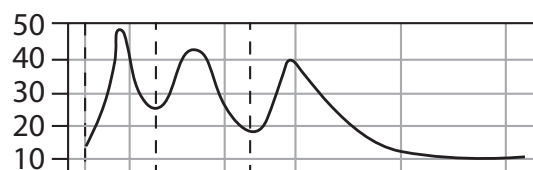
Blood glucagon concentration  
/ng cm<sup>-3</sup>



Blood glucose concentration  
/mg 100 cm<sup>-3</sup>



Blood insulin concentration  
/units cm<sup>-3</sup>



8 am    Noon    4 pm    8 pm    2 am    8 am  
▲    ▲    ▲  
meal meal meal

Discuss how negative feedback is involved in the control of blood glucose concentrations.

Use the information in the diagram and the graphs to support your answer.

(6)

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(Total for Question 7 = 15 marks)